



Empowering facial emotion recognition in service industry – a two-stage convolutional neural network model

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Abstract

Deep-learning based facial emotion recognition (FER) has potential in the service industry. To solve conventional emotion recognition problems, this study proposes a deep learning-based model to achieve a highly-efficient FER system. The proposed FER model consists of two stages. The first stage uses the multitask convolution neural network to distinguish precise face-bounding box positions, whereas the second adopts a deep-learning network to achieve real-time recognition of emotions and features. By training our model using three global FER datasets, its accuracy indicates that the proposed model outperforms existing FER models. The study illustrated the model from three aspects. First, a massive facial database is investigated for model feasibility with a variety of service scenarios. Secondly, we demonstrated practical examples in the restaurant and retailing service industries. Third, the model performs advice by monitoring the emotion when the player is assembling Lego. The model can analyze human emotions in the service industry to identify customer satisfaction with products and/or services, fatigue in working domains, and/or safety in jobs. The model will help improve customer relationships, provide pleasant transaction solutions, and even help to broaden product offerings and promotions. Such facial recognition technology can further motivate new digital business models and change customer-server dynamics.

Keywords Convolution neural network · Deep learning · Emotional intelligence · Facial emotion recognition · Service industry

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1 Introduction

Emotional intelligence emerges to accelerate new digital transformation in business [1]. With the development of deep learning, facial emotion recognition (FER) has become widely used in different industries, such as user experience [2], human–computer collaboration [3], and medical diagnosis [4]. In the past decade, artificial intelligence (AI) based FER has gained the interest and attention of researchers in the service field [5]. Good human communication relies on identifying facial expressions, and this process involves emotional computing [6].

The FER can efficiently identify humans' intentions, feelings, and cognitive states; hence, promotions and advertisements can be targeted according to these manifestations. However, conventional emotion recognition strategies supported by image process technology suffer from low recognition accuracy and problems with real-time feature extraction. In addition, recognition accuracy is easily harmed by blank and/or blurry images in a video [7].

Across the world, people show similar expressions of the same emotion. Ekman & Friesen [8] categorized facial emotions into six categories, namely, happiness, fear, sadness, surprise, disgust, and aggression. These emotional states can be distinguished from neutral states. FER tasks can be performed by supervised learning from massive datasets [9]. In computational FER methods, manually performing the labeling and selection of feature extraction is necessary, and these tasks not only require substantial manpower work; the accuracy of feature extraction is also influenced by subjective judgment [10].

In this article, we developed a deep-learning based a two-stage FER system. In the first stage, the system performs automatic alignment and marking of faces in the dataset. The images were then input into the second stage for FER. We combine the multi-task cascaded convolutional neural network (MTCNN) [11] and the You Only Look Once (YOLO) v3 algorithm [12], in which MTCNN aligns and detects human faces, and YOLOv3 conducts FER in real-time.

The contribution of this study lies in two aspects. First, this study proposed a deep-learning based two-stage FER system that can effectively detect emotions. Second, the proposed model can be deployed in real-time to analyze human emotions in the service industry to identify customer satisfaction with products and/or services, fatigue in working domains, and safety in jobs. Such facial recognition technology can motivate new digital business models and change customer-server dynamics.

The rest of this paper is organized as follows. Section 2 summarizes previous research and explains MTCNN and YOLOv3 structures as well as FER. Section 3 provides our framework and method. Section 4 demonstrates the experiment results. Lastly, Section 5 summarizes our research.

2 Literature survey

2.1 Facial expressions and emotions in service industry

Designed to chat or observe like people, AI is ready to deploy in a frontline in service processes [13]. AI for capturing and sensing the emotion of customers and employees will facilitate prompt and premium service interactions. On one hand, the detection intelligence of customer

emotion is a moderator between service activities and customer reaction and has a significant effect on customer engagement [14, 15]. On the other hand, employees monitored by AI (through employee emotion detection) significantly regulate employee service quality and affect customer emotions subsequently. Augmenting service employees with AI becomes critical [16].

Human expression standards are mainly based on the facial action coding system (FACS) [17]. FACS defines 44 different action units (AU) by using the principle that the face will drive different facial muscles when showing different expressions. Therefore, various expressions of the human face show different AU combinations. AU derived in feature extraction is a dynamic feature; that is, several AUs can be detected from the beginning to the completion of an expression action. Usually, the number of AUs from the beginning of an expression action to the apex (the largest change) is taken as a representative. Among the defined AUs [18], 30 AUs are anatomically related to the contraction of specific facial muscles: 12 are utilized for the upper face, 18 are utilized for the lower face, and AU 1 to 7 address the eyebrows, forehead, or eyelids. Based on specific facial expressions, scholars proposed a basic expression classification, including angry, disgust, surprise happiness, sadness, and fear, in addition to a neutral emotion. Advances in emotional theories have occurred over the past decades, and those research in marketing about emotions has exploded. A comprehensive review of emotions, including their relation to decision-making and action, as well as measurement issues, can be seen in [19].

FER is used in service sectors, health care, medical applications, and self-driving cars. Facial expressions play a crucial role in recognizing people's emotions to determine their intentions, feelings, and cognitive states [20]. Research showed that 7% of communication information is transmitted through language, 38% is transmitted through paralanguage, and 55% is transmitted through facial expressions [21]. FER consists of three procedures, which are face unit detection, feature extraction, and facial emotion classification [22] Fig. 1 shows an example of a facial expression recognition pipeline.

2.2 Deep-learning based FER

Convolutional neural networks (CNN) were one of the foremost productive deep models for image detection and analysis. CNN has been extensively used in drivers' computer vision applications to correct face scale variation and alignment [23]. In the process of CNN training, a common problem is uneven training data. Hence, a data enhancement method, such as horizontal flip and random cropping, is proposed to solve the problem of insufficient labeled samples. In addition, numerous pre-processing strategies have been extensively utilized to hold expression-associated features in facial images [24]. Kim et al. [25] trained and combined multiple CNN models to tackle the FER issue. The feature position of FER was learned by R-CNN. By generating high-quality regional proposals, faster R-CNN is utilized to recognize facial emotions [24].

YOLO and MTCNN are specializations of CNN, but serve different functions in the FER study. YOLO series is a state-of-the-art one-stage objection system. YOLO is used for object detection and classification, in which it is mainly composed of 5 maximum pooling

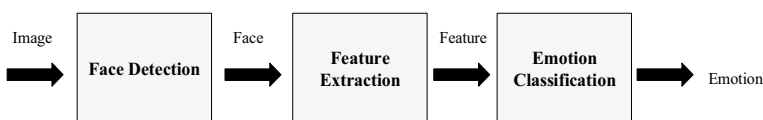


Fig. 1 FER pipeline

layers, 19 convolutional layers, and fully connected pooling layers. The original YOLO network has been modified and updated to YOLOv2 by using the multi-scale training method, running in different sizes with keeping speed and accuracy in good balance. YOLOv3 was released in 2018. The algorithm has a Darknet-53 network that can be used to obtain features. Huang et al. [26] developed YOLO-LITE, which runs on portable devices (such as laptops) without GPU. Recently, YOLOv3 has been utilized in FER [10, 28]. YOLOv3 is applied to JAFFE, RAF-DB, and CK+ databases to evaluate FER performance and effectiveness [9]. However, only a few literature discussed how pre-processing data sets before using YOLOv3 could benefit the accuracy of FER.

YOLOv3 has been widely used in object recognition and object classification [27]. Compared with other YOLO algorithms, it has the following characteristics [12]. (1) It processes at three different levels and subsequently adds three up-sampling layers. (2) It improves the detection of small objects but worsens that of large object predictions. (3) It uses nine anchor box models for bounding box prediction with more bounding boxes per image. (4) It uses logistic regression to predict objects; that is, it replaced the cross-square error with the cross-entropy loss function. However, Yolo is a generic object detection tool but not dedicated to facial detection. It also suffers from the problem in which flawed images of a face cannot be detected in the training dataset and the problem of labeling images taking too much time.

Moreover, the Cascade Haar-like face detector [29] has been used for traditional FER tasks, but face vision changes greatly, and its application possibilities are limited. Inspired by the success of CNN in computer vision applications, Zhang et al. [30] proposed an MTCNN face detector. MTCNN is a deeply cascaded network structure that uses internal cascading between each other to improve the overall network performance. It is mainly utilized in detecting and aligning faces in images. The feature of MTCNN is that it recognizes a face by locating the features of the face, detecting the area of the face, and aligning the face as well. Therefore, the MTCNN structure is composed of an image pyramid connected with three sub-deep learning networks, which are Proposal Net (P-Net), Refine Net (R-Net), and Output Net (O-Net). The details of MTCNN and YOLO refer to online appendixes 1 and 2.

Emotional intelligence is also implemented by several world-leading commercial solution providers. Google Cloud Vision and Amazon Rekognition provide a large scope of FER functions [31]. Google Cloud Vision can detect joy, sorrow, anger, and surprising emotion. Amazon Rekognition can detect happy, sad, angry, confused, disgusted, surprised, and calm. The emotional set chosen by Amazon is almost identical to the universal emotions. Amazon Rekognition performs better than Google Vision. However, a test on platforms' performance for no-wrong emotion detection only reaches 36% and 64% by Google Vision and Amazon Rekognition respectively.

Previous studies compared the Face ID, Microsoft Azure Face API, Google Cloud Vision API, and Amazon Rekognition Image [32]. Azure Face API can detect 6 human emotions — happiness, sadness, fear, contempt, surprise and anger, and neutral emotion. Their study focused on 4 criteria: accuracy, service processing speed, recognition cost for each function, and additional features and recognition capabilities. Face ID reaches a rapid response time of 0.77 in seconds but without an emotion detection function.

It is worth noting that these commercial FER solutions require intensive investment for industry usages and are unaffordable to many small-to-medium enterprises, in addition to their encrypted algorithmic details as a barrier for further improvement, modification, and customization.

2.3 Summary

Over the past several years, supervised learning has gained popularity in the field of emotion recognition. Conventional CNN frameworks cannot fulfill the speed requirement of real-time FER and require multiple labeled images in the training process. Marking images wastes much time and is too subjective. Improving accuracy is another issue suffered by current commercial face emotion solutions. Therefore, establishing a hybrid real-time FER model is required. MTCNN is capable of aligning and detecting face images in the dataset. YOLOv3 is efficient in extracting features and achieving real-time FER. Therefore, how to integrate the two advantages is a key issue.

3 Method

3.1 Research framework

This study proposes a unified model that integrates two different CNNs. We developed a deep-learning based two-stage FER system that can effectively detect emotions to be consistent with Ekman's six basic emotions (i.e., anger, surprise, disgust, enjoyment, fear, and sadness). Figure 2 shows the framework for the proposed model. This model benefits from the two individual modules. An MTCNN is employed to properly align the face boundary. The other purpose of using MTCNN is to remove blemishes on images. Then, it conducts the pre-processing of alignment and automatic labeling of images in the dataset. Online appendix 3 shows the illustration of defective images. MTCNN can detect and align faces in images, and frame the position of faces by detecting the position of the eyes, nose, and corners of the mouth. Then, YOLOv3 is used to detect emotions with the best depth in real time. The automatically marked emotional classification images are obtained.

Figure 3 shows the entire process of the two-stage model. The training configuration of YOLOv3 defines the parameters needed for the training, evaluation, and inference. Details refer to NVIDIA [51].

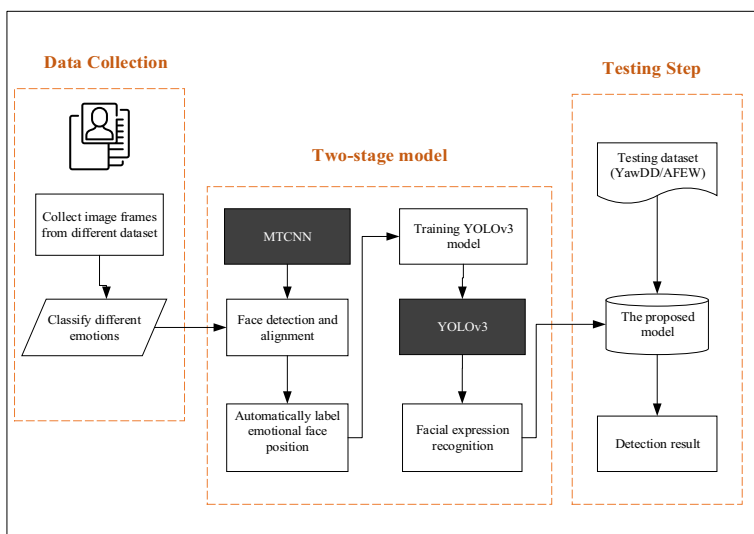


Fig. 2 Framework of the proposed two-stage FER model

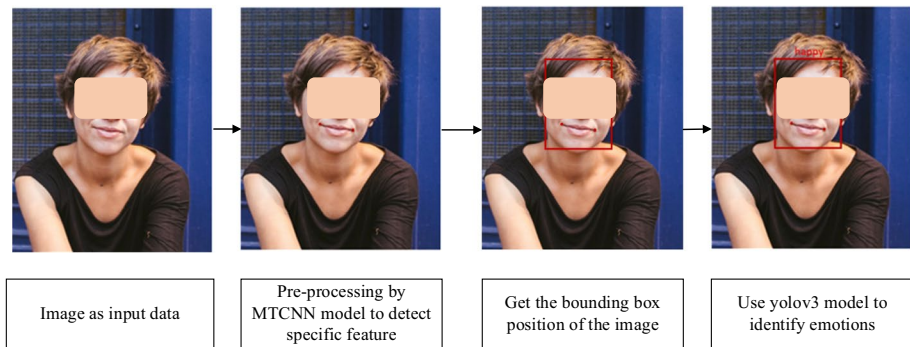


Fig. 3 The proposed model scheme

The algorithm of the proposed two-stage FER model is presented as follows.

Input: Images of database

Output: FER-trained model

Parameters:

Batch size — the size of samples of network

Target size — image size after resized

Activation — network layer activation function

Learning rate — determines the speed of weight update

Max batches — the number of times the training loop update parameters

Anchor — the initial width and height of the prediction box

BEGIN

Step 1: Construct pre-trained MTCNN

- Define the train data based on the emotion categories

Step 2: Database preprocessing

- Eliminate images where no face can be found
- Automatically label emotional face position

Step 3: Construct the YOLOv3 architecture

- Set input shape and learning rate
- Set activation function
- Set anchor comparison

Step 4: Set the model optimizer (explained in Table 2)

Step 5: Examine the optimal parameter setting to get the best model

- Save the best YOLOv3 trained model

Step 6: Validate and test model

- Use video and dataset to test

END

3.2 Facial expression databases

FER studies commonly use open-source databases for model training and validation, such as FER2013 [33], real-world affective faces (RAF) [34], and extended Cohn-Kanade (CK+) [35]. The three databases (refer to online appendix 4) and their combination are used in this research and investigated in descending order of their number of images.

The mixed dataset is a dataset combining FER2013, RAF-DB, and CK+ datasets. We use this mixed data set to increase the size and the diversity of images, exploring whether combining multiple images can help improve accuracy.

3.3 Performance indicator

The confusion matrix is used as a model verification indicator. The FER results are divided into true positive (TP), false positive (FP), true negative (TN), and false negative (FN). Table 1 shows the confusion matrix of the FER results.

To compare our model with the other models, we measure the classification performance in terms of precision, mAP, recall, and F1-score. Precision defined as Eq. (1) means that the proportion of identification was correct; recall in Eq. (2) means the proportion of actual positives was identified correctly; F1-score in Eq. (3) is a metric that considers precision and recall; mAP in Eq. (4) means the average precision, which is obtained by accumulating the mean ratio of the correct objects in the images, where n_{cls} is the number of categories and AP_i is the average precision of each class.

$$precision = \frac{TP}{TP + FP} \quad (1)$$

$$recall = \frac{TP}{TP + FN} \quad (2)$$

$$F1 - score = 2 \times \frac{(recall \times precision)}{(recall + precision)} \quad (3)$$

$$mAP = \frac{\sum_i^{n_{cls}} AP_i}{n_{cls}} \quad (4)$$

3.4 MTCNN for pre-processing

In this study, implementation of the MTCNN face detector is done using Keras in Python3.4+, using an MTCNN implementation reference from (FaceNet's MTCNN) [30]. The MTCNN includes four kinds of data annotations in the training model for executing face detection and alignment simultaneously, which are negatives, positives, partial faces, and landmark faces. The Intersection-over-Union (IoU) ratio of a region compared to a ground truth face less than 0.3 is marked as a negative. Positives mean that the IoU ratio

Table 1 Confusion matrix for evaluation

Labeled	Predicted	Confusion matrix
Positive	Positive	TP
Positive	Negative	FP
Negative	Positive	FN
Negative	Negative	TN

of a region is larger than 0.65. If the IoU ratio is between 0.4 and 0.65, then the region will be labeled as a facial face. Faces with five landmarks' positions are classified as landmark faces. These four annotations make a significant contribution to subsequent programming. The face classification tasks require negatives and positives, and the bounding box regression takes positives and facial faces as parameters. The landmark faces are applied for facial landmark localization.

At the first stage of the proposed FER model, we use MTCNN in which the face detector automatically crops faces in the image. The detector returns a list of JSON objects [30]. Online appendix 5 shows the MTCNN face detection procedure. Images that cannot be detected by MTCNN will be deleted. Given that the images mainly captured by the MTCNN model need to have five feature points: eyes, nose, and mouth, images with the head slightly tilted up or toward the side are easily deleted. We select the data in the box in the JSON file, convert them to a txt file and add emotion categories (0 = surprised, 1 = fear, 2 = disgust, 3 = happiness, 4 = sadness, 5 = anger, 6 = neutral). The image type and the bounding box are established into a txt dataset. Finally, they reach the second stage of the proposed FER model for model training by using YOLOv3.

The output format is represented as JSON, consisting of three primary keys: box, confidence, and key points (the detail refers to Fig. A5–2 in online Appendix 5). The boundary box is organized as [x, y, width, height] under the box. Confidence is the probability that the bounding box matches the face. The key purposes are formatted into a JSON object with the left eye, right eye, nose, left corner of the mouth, and right corner of the mouth. Every key point is known by an element position (x, y).

3.5 YOLOv3 for detecting emotions

The implementation steps of YOLOv3 are as follows. (1) Clone repository: We cloned the Darknet53 repository from GitHub by AlexeyAB [36]. (2) We built the training and testing set. Specifically, we prepared “train.txt” and “test.txt” files that contain the label coordinates of the training and validation images. (3) We edited “obj. Data,” “obj. Names,” and “class.txt.” The file “obj. Data” will contain the number of classes, and the “class.txt” file contains all the training data classes. (4) Set-up of the configuration file: This step is one of the most important steps when setting up YOLOv3, because it defines how to train the network. Table 2 explains in detail each main parameter. (5) Training: To significantly reduce training time, we use transfer learning from given pre-training weights provided in the Darknet GitHub repository. Training 2000 images took approximately 4 hours. (6) Uploading of the validation data set. (7) Validation: We used the validation dataset to calculate the accuracy of the CNN and obtained a medium average

accuracy (mAP). (8) Testing: Tests were run using open-source image databases to demonstrate the detections performed.

The configuration parameters are explained to better understand the network, as shown in Table 2(a). During the training process, the model is optimized by fine-tuning the parameters. Table 2(b) shows the performance of fine-tuning.

The training is completed by using the *k*-fold approach. Darknet outputs a graph that illustrates how the loss function drops in each iteration. The training process was designed through 14,000 iterations of the YOLOv3 experiments. The accuracy of each network was presented every 2000 iterations. Figure 4 indicates that the training accuracy curve becomes steady when the epochs approach 2000. We fine-tune the YOLOv3 parameters by referring to previous research [36–38]. First, we set the batch size to 64, momentum to 0.9, and weight decay to 0.0005 according to a normal objective detection approach. Then we adjust different input resolution sizes and learning rates by a greedy search to compare mAP, loss value, and training time. Finally, the best outcome suggests an input resolution size of 480×480 and a learning rate set at 0.001.

The Darknet-53 network of YOLOv3 is constructed by using the structure of the ResNet residual network, and it sets up 13×13 , 26×26 , and 52×52 three different sizes of anchor boxes to extract image features. The smaller the anchor boxes used; the more details of the target object can be obtained. Owing to subtle changes in facial emotions, YOLOv3 can quickly capture subtle expression changes. After the training phase is completed, YOLO will generate a trained model and prepare for facial expression recognition testing. We will use the trained YOLO model to detect and classify the facial expression categories on each test image, and generate a bounding box containing the location, classification results, and confidential information.

Figure 5 shows the overall flow chart of the FER algorithm designed in this paper. First, the video is input into the algorithm, and then the video is decoded into a picture. Then, each frame of the image is sent to the YOLO v3 network to identify the type of emotional expression.

4 Model evaluation and discussion

The hardware and operating system used in the model evaluation consists of Windows 10 OS, Intel(R) Core(TM)i7-9700F CPU, NVIDIA Geforce RTX 2070 SUPER GPU, 48 GB memory, 2 TB SSD storage, and the parallel computing framework of CUDA v10.2, cuDNN v7.6.5, OpenCV v4.2.0.

4.1 MTCNN evaluation

We train the models on three different datasets. First, we use the MTCNN model to pre-process the FER2013, RAF-DB, and CK+ datasets to filter out unsuitable images in the training and testing phases. After calculating the ratio of each expression in the dataset and deleting flawed images, similarly distributed training and test sets can be retained. The distribution in the table refers to the ratio of the number of each emoticon picture in the number of all pictures in training and testing. Table 3 shows the distribution of each class.

The result indicates that after the data pre-processing by MTCNN, the total number of images in FER2013 is reduced by 30%, RAF by approximately 15%, and CK+ by 21%. FER2013 has reduced the number of images due to more flaws.

Table 2 YOLOv3 configuration parameters

(a) Hyperparameters	
Parameters	Description
<i>#Training</i> batch = 64 subdivisions = 32 width = 480 height = 480 channels = 3 momentum = 0.9 decay = 0.0005 learning rate = 0.001 max batches = 14,000	Before training, the study first resized the input training image to width x height. If one increases the input resolution to 608 × 608, the result may be improved, but the training time will be significantly long. Therefore this study adopts the default values of 480 × 480. channels = 3 means that we would process 3-channel RGB input images. The value of momentum and decay determines how the weight is updated. Update the CNN weights based on a small batch of images instead of the entire dataset. The weight update fluctuates, so we use the parameter momentum to punish giant weight changes between iterations. CNN can easily be overfitting during training, and the way to mitigate this problem is to use parameter decay to penalize larger weight values. The parameter learning rate controls how aggressively we learn in batches based on the current data. Usually, this number is between 0.01 and 0.0001. We need to run more batches in a multi-class object detector. For n-type object detectors, it is recommended to train at least 2000 * n batches.
(b) Performance result by fine-tuning parameters in training	
mAP	Learning rate 0.0001 0.0001
Input size	416×416 85.3% (loss: 0.037) Run time: 12 hr 480×480 85.8% (loss: 0.0396) Run time: 12 hr 82.9% (loss: 0.0827) Run time: 12 hr 85.3% (loss: 0.0903) Run time: 14 hr

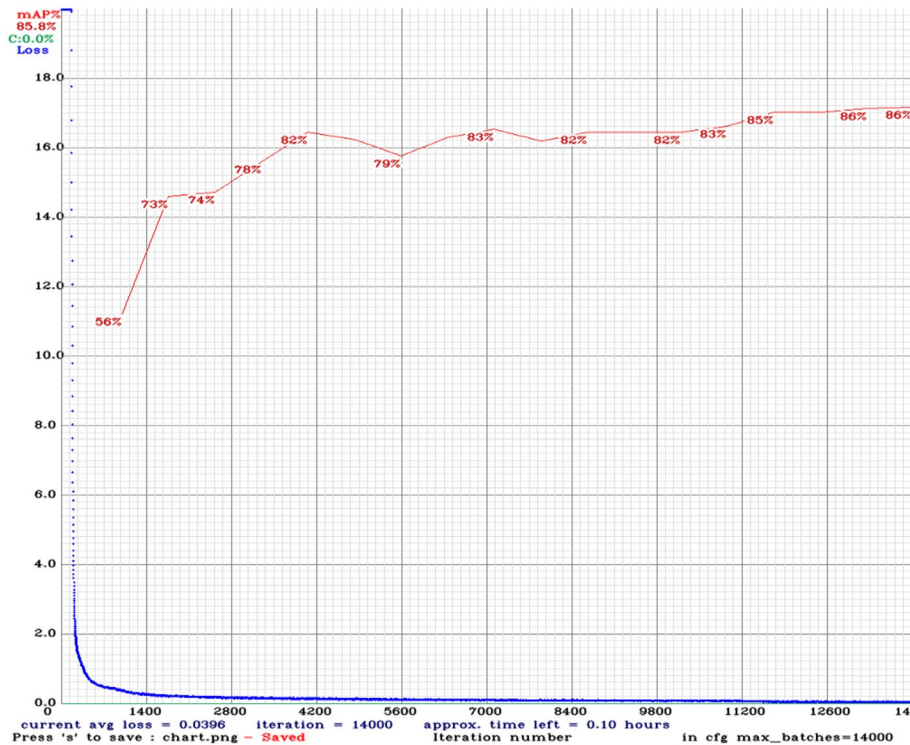


Fig. 4 Training performance of the YOLOv3 model for emotion detection

4.2 The proposed FER model evaluation

We compare the YOLOv3 model with the proposed two-stage FER model. We selected 70% of the data for training, 20% of the data for verification, and the remaining 10% for the test in each dataset. Table 4 shows the comparison among FER2013, RAF-DB, CK+, and a mixed database with the above three databases. Results indicate that the proposed FER model performs particularly well in detecting emotions of grayscale, flawed and low-pixel facial images. It improves the precision and F1-Score in most of the databases. Using MTCNN eliminates background noise, which lets the position of the bonding box become accurate and improves model precision.

The precision of the model increased from 64% to 68% by using the FER2013 dataset that the overall images are black and white and the pixels are low. Further, both the proposed model and YOLO model perform well in the RAF-DB dataset. Because the RAF-DB dataset has fewer flawed images and the use of the MTCNN model reduces the number of images for training, the performance of the proposed model is slightly reduced. In the CK+ dataset, the proposed model remains the same performance as compared to the YOLOv3 model. The proposed model has better precision than that with only YOLOv3 in testing the mixed database because half of the mixed database is from FER2013.

Moreover, when applying the proposed FER model to the yawning behavior of car drivers, it reaches an accuracy of 82% which is effective in practical use (refer to online appendix 6).

Fig. 5 Flowchart of facial expression detection algorithm

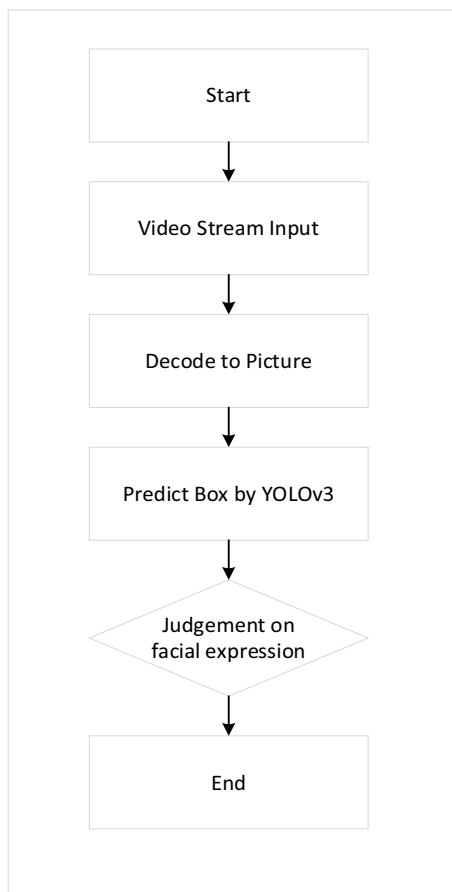


Table 5 shows AP in different databases and methods. The overall mAP of the proposed model ranges from 69% to 96% in the three facial datasets and outperforms those (67.1 to 95.6%) by the YOLO model, as well as those (36% to 64%) by Google Vision and Amazon Rekognition in literature [31]. Specifically, considering that the overall images in the FER2013 dataset are grayscale and the pixels are low, the overall mAP reaches 69%.

In terms of emotion types, happiness and surprise have the two highest average precision because they have distinct facial features such as an open mouth or a smile with a mouth curled upward. In addition, the emotions of disgust and fear result in lower average precision because of invisible changes in facial expression. These negative emotions may also share similar features such as frowning, corners of the mouth curled downward, and pressed lips. These similarities are difficult to distinguish.

Moreover, we qualitatively evaluate different object detection models based on their characters in the specific domain of the present study. Although YOLOv4 and YOLOv5 are generally considered an improvement over YOLOv3, with better accuracy and faster training and inference times. However, YOLOv3 is preferable in the present study owing to the following considerations where YOLOv3 is a better choice: (1) Hardware limitations: YOLOv4 and YOLOv5 require more powerful hardware to train and run efficiently

Table 3 The distribution of emotion classes

(a) FER2013 dataset: Number of images per class								
	Surprise	Fear	Disgust	Happy	Sad	Angry	Neutral	Total
Training	3171	4097	436	7215	4830	3995	4965	28,709
Testing	277	352	37	586	396	328	418	2394
Validation	139	176	18	293	198	163	208	1195
Distribution	11%	14%	2%	25%	17%	14%	17%	100%
Training (MTCNN)	2361	2549	339	5545	2636	2748	3706	19,884
Testing (MTCNN)	204	228	29	461	202	223	303	1650
Validation (MTCNN)	101	114	15	230	100	112	152	824
Distribution (MTCNN)	12%	13%	2%	27%	13%	14%	19%	100%
(b) RAF-DB dataset: Number of images per class								
	Surprise	Fear	Disgust	Happy	Sad	Angry	Neutral	Total
Training	1290	281	717	4772	1982	705	2524	12,271
Testing	219	49	107	790	319	108	453	2045
Validation	110	25	53	395	159	54	227	1023
Distribution	11%	2%	6%	39%	16%	6%	21%	100%
Training (MTCNN)	1048	228	660	4278	1601	583	1997	10,395
Testing (MTCNN)	178	41	96	701	246	91	369	1722
Validation (MTCNN)	89	20	48	351	123	45	184	860
Distribution (MTCNN)	10%	2%	6%	41%	15%	6%	20%	100%
(c) CK+ dataset: Number of images per class								
	Surprise	Fear	Disgust	Happy	Sad	Angry	Neutral	Total
Training	166	25	59	69	28	45	593	1003
Testing	33	8	12	14	6	9	119	201
Validation	17	3	6	7	3	5	59	100
Distribution	16%	2%	5%	9%	4%	5%	59%	100%
Training (MTCNN)	166	25	59	69	28	45	593	1003
Testing (MTCNN)	33	8	12	14	6	9	119	201
Validation (MTCNN)	17	3	6	7	3	5	59	100
Distribution (MTCNN)	16%	2%	5%	9%	4%	5%	59%	100%

than YOLOv3. Since we do not have access to high-end GPUs or specialized hardware like Google's TPU, YOLOv3 becomes a more practical choice. (2) Small object detection: Our application requires detecting very small objects, YOLOv3 may still be the

Table 4 The training results by the proposed two-stage FER method in different datasets

Database	Method	Precision (%)	Recall (%)	F1-Score (%)	Computational Training Time (hr)	Computational Testing Time (sec)
FER2013	YOLOv3	64	75	66	14	0.022
	MTCNN+YOLOv3	68	78	68	14	0.024
RAF-DB	YOLOv3	84	90	86	14	0.032
	MTCNN+YOLOv3	81	88	84	14	0.034
CK+	YOLOv3	96	97	97	12	0.022
	MTCNN+YOLOv3	97	97	97	12	0.024
Mixed Database	YOLOv3	74	81	73	16	0.028
	MTCNN+YOLOv3	76	78	72	16	0.030

better option. (3) Legacy code: We have successfully implemented YOLOv3 in our code-base and it is working well for our research needs and in the computing platform of the case study company, there is little reason to upgrade to YOLOv4 since you need no other specific features or improvements, and (4) Limited training data: YOLOv5 requires a large amount of training data to perform well. Because we have limited training data available, YOLOv3 is still a better choice as it can perform well with smaller datasets.

Further, YOLOv3 and Vision Transformer (ViT) [52] have different strengths and weaknesses depending on the task at hand. For example, YOLOv3 is typically faster than ViT at detecting objects within an image, especially when there are multiple objects or when the objects are small. The situation is exactly the issue we addressed in the study. This is because YOLOv3 uses a single-stage detector that performs object detection and classification in a single forward pass. On the other hand, ViT is typically more accurate than YOLOv3 at classifying images into different categories. This is because ViT uses a transformer-based architecture that is designed to capture long-range dependencies between different parts of an image.

5 Applications to the service industry and discussion

5.1 The proposed FER model applications to acted facial expressions in the wild (AFEW) dataset

To demonstrate the usability of the proposed model in different facial emotion, a database that imitates human emotion in the real world are investigated. Acted Facial Expressions in the Wild (AFEW) [39, 40] is a dynamic emotional faces dataset consisting of a close to real-world environment collected from famous Hollywood movies. The dataset is divided

Table 5 Comparison of different emotions on AP

Database	Method	AP (%)							
		Happy	Neutral	Sad	Surprise	Disgust	Angry	Fear	mAP
FER 2013	YOLOv3	93.1	71.0	56.2	83.6	50.7	62.5	52.6	67.1
	MTCNN+ YOLOv3	94.1	71.7	48.8	86.1	63.8	63.6	54.6	69.0
RAF-DB	YOLOv3	97.4	89.2	89.9	89.9	75.7	93.4	67.1	86.1
	MTCNN+ YOLOv3	95.2	90.4	86.2	86.2	65.8	77.2	67.6	81.4
CK+	YOLOv3	100	98.9	66.7	100	100	100	100	95.6
	MTCNN+ YOLOv3	100	93.2	86.9	100	93.2	100	96.6	96.0
Mixed Database	YOLOv3	86.2	92.8	68	81.8	70.3	65.6	55.7	74.2
	MTCNN+ YOLOv3	94.2	80.6	73.7	87.1	69.6	69.6	55.1	75.7
	YOLOv3	94.2	88.0	70.2	88.8	74.2	80.4	68.9	80.8
Average	MTCNN+ YOLOv3	95.9	84.0	73.9	89.9	73.1	77.6	68.5	80.5

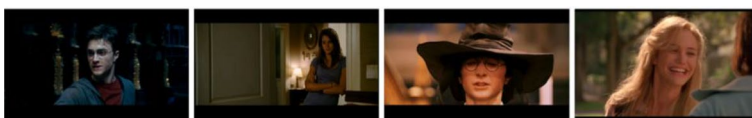
into seven expressions (happiness, disgust, fear, anger, neutral, surprise, and sadness), which are the same types of expressions as the training dataset we use. The AFEW set-up contains 711 training films, 539 test films, and 371 verification films. We use the validation data set in this experiment. Figure 6 shows the samples of the AFEW, including the illustrations of seven emotions [41] and sample emotions with different scenarios.

We select 10 service-related videos from each expression category, containing scenarios such as restaurant service, party activities, conversation and greetings, driving-thru service, military training and commends, medical diagnostics, street advertising, museum tutoring, nursing, and quarrel. The video is cropped in 25 frames per second, and each frame is input into the model test and the confusion matrix is used as the result. Table 6 shows a comparison of the precision, recall and F1-Score of the proposed model. The proposed model improves by 7.8% in F1-Score as compared to Yolov3.

We use confusion matrices to compare the predicted emotions with true emotions from the database. In this paper, we use the normalized confusion matrix to present training results. If a number in the normalized confusion matrix is closer to 1, then the predicted results better match the true results. Figure 7 shows the accuracy of anger and neutral is higher than that of other emotions.

5.2 The proposed FER model applications to the restaurant and retailing service industry

Customer emotion and employee emotional labor result in significant impacts on service encounters [28]. We illustrated several examples in the service industry [42–44], and analyzed the emotional responses between customers and waiters by our models. These videos were annotated by two experts in facial expression and sentiment analysis. If two experts reach a consensus on the video, then the video will have the emotions agreed upon by both parties. The two-stage CNN model using YOLOv3 trained on the RAF-DB dataset is employed to test the videos.



(a) Sample emotions in the AFEW dataset with different scenarios

Natural		Fear	
Surprise		Angry	
Sad		Disgust	
Happy			

(b) The seven emotions of the AFEW dataset (revised from [41])

Fig. 6 Samples of facial expression data in AFEW

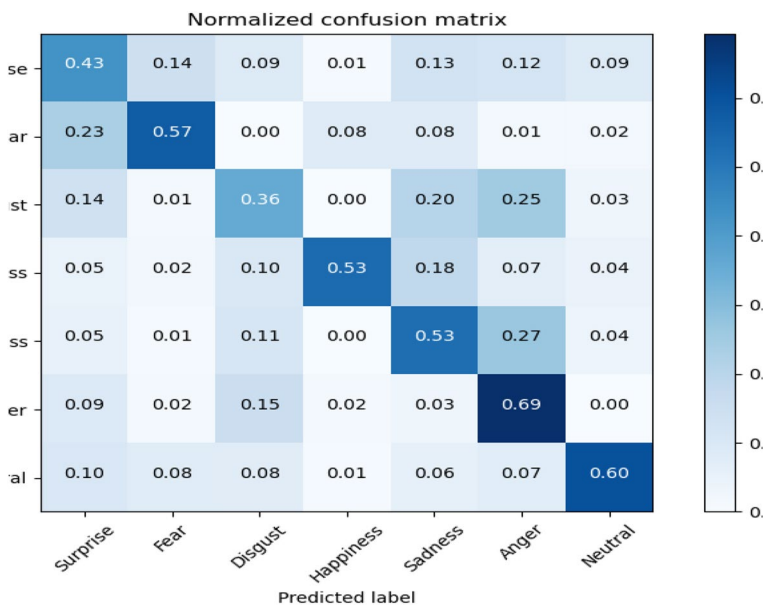
Table 6 Precision and recall of different methods on the AFEW dataset

	Precision (%)	Recall (%)	F1-Score(%)
YOLOv3 ^(a)	56.6	53.2	53.3
MTCNN+YOLOv3 ^(b)	62.5	60.9	61.1
Improvement (%) ^{(b)-(a)}	5.9%	7.7%	7.8%

In terms of emotion, we collected two videos, each spanning 15 seconds. The employee in video 1 has 303 frames and the customer in video 2 has 375 frames. Figure 8 shows an example of video detection results. In the first video, a fast-food restaurant employee shows his enthusiasm for service, providing friendly and ethical assistance to customers. The expression of happiness is largely detected due to the expression of massive smile emotions. The service quality attributes, customer satisfaction, and trust affect customer loyalty to the fast-food restaurant [45]. By the FER model to detect customer emotions in the service process, customer loyalty can be optimized.

In the second video regarding retailing activities, the customer berates the clerk because of personal emotional factors. The service quality of the retailer influences legitimacy, resulting in customer revisit intention [46]. Throughout the emotion recognition process, consumers' emotions can be effectively captured and utilized to improve their on-site service experiences.

The results in Table 7 indicate that the FER on the two videos detects more frames in the two-stage model than in the YOLOv3 model.

**Fig. 7** Normalized confusion matrix

5.3 A case study of the proposed FER model on Lego players

5.3.1 Emotion collection

The emotion dataset was obtained through a video recording of a Lego assembly process. The Lego players were asked to perform the assembly following a specified procedure while being recorded. In the experiment, we recorded 15 videos. Then, each video was extracted into consecutive images with a rate of 30 video frames per second. The dataset consists of 35,887 Gy-scaled images of faces with a size of 48×48 . They were categorized into seven types of emotions: anger=0, disgust=1, fear=2, happy=3, sad=4, surprise=5, and neutral=6. Moreover, 80% of the data was used for training, and the remainder was used for validation. The distribution of the emotion dataset in this experiment can be found in Fig. 9.

The players' motion was classified into five types. The heart rate of the player is also collected simultaneously. The normal range of heart rate as suggested by the American Heart Association is 60–100 beats per minute (BPM) [47].

5.3.2 Training and validating the FER model

The training results of the FER model are summarized in Fig. 10. The training and validating loss are around 1 while the training and validating accuracy are around 68%, which is competitive to recent benchmarks of 58.8% in the human machine collaboration (HMC) domain [41].

5.3.3 Result

The FER model continuously scores the state of the player by monitoring the emotion, along with additional motion and heart rate monitoring functions. Accordingly, appropriate recommendations will be provided. We conducted three example cases in which the exception level increases to illustrate the proposed model. The first case (Fig. 11a) presents the normal playing state. If the player is performing the right Lego assembly motions with a neutral emotion and heart rate, the model will provide a “keep working” recommendation. In the second case (Fig. 11b), if either the negative emotion happens, the model will

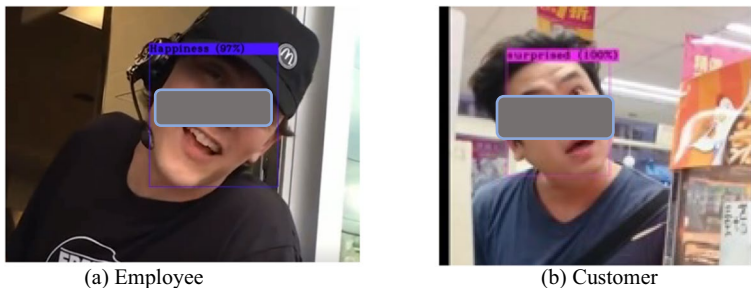


Fig. 8 Example video detection result

Table 7 Employee/customer emotion detection by different methods

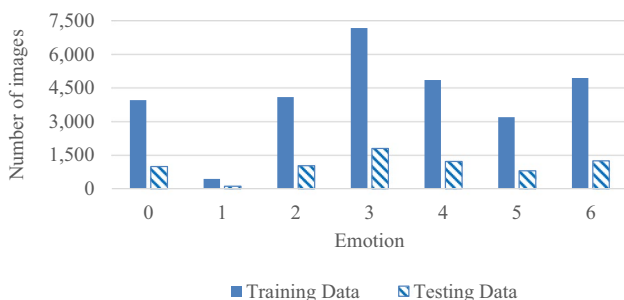
Video	Method	Number of frames								mAP(%)
		Happy	Neutral	Sad	Surprise	Disgust	Angry	Fear	Total	
a	YOLOv3	190	43	0	35	5	0	0	273	74
	Expert count	260	13	0	0	0	0	0	273	
	MTCNN+	194	55	0	42	0	0	0	291	85
	YOLOv3									
b	Expert count	267	24	0	0	0	0	0	291	
	YOLOv3	0	30	23	153	83	43	0	332	42
	Expert count	0	0	0	72	23	237	0	332	
	MTCNN+	0	25	20	186	121	21	0	373	42
	YOLOv3									
	Expert count	0	0	0	83	51	239	0	373	

remind the player to keep concentrating on the assembly process. The last case (Fig. 11c) presents the high-risk level when the heart rate is abnormal or fear emotion happens. In this case, the model will alarm to stop playing and the player will have time for a health check and/or relaxation.

This application model also provides a safety-enabling function for the HMC that a human operator has shared tasks and workplace with a machine [1]. The model can monitor the emotions, motions, and vital signals of the human operator continuously. If abnormalities happen, appropriate guidance or actions can be provided by the model.

5.4 Discussion

The study has illustrated the proposed FER model from three aspects. First, to demonstrate the usability of the proposed model in different facial emotions, a massive facial database (i.e., AFEW) that imitates human emotion in the real world is investigated for the illustrations of our model feasibility in seven emotions with a variety of service scenarios, such as restaurant service, party activity, driving-thru service, military training and commend,

**Fig. 9** Distribution of the emotion dataset recorded in a Lego playing process

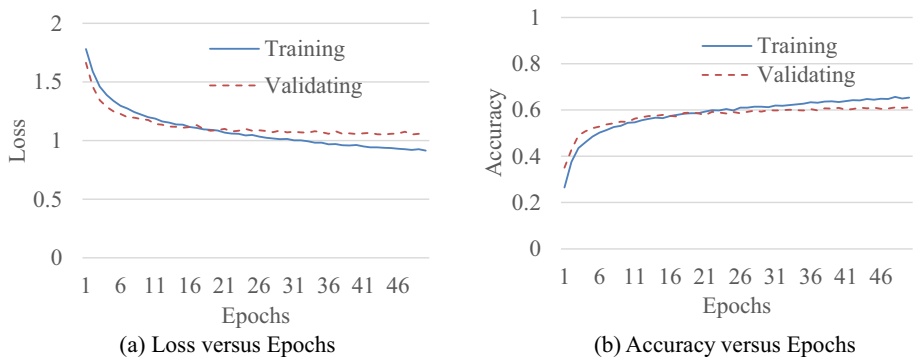


Fig. 10 Loss and accuracy during training the proposed FER model

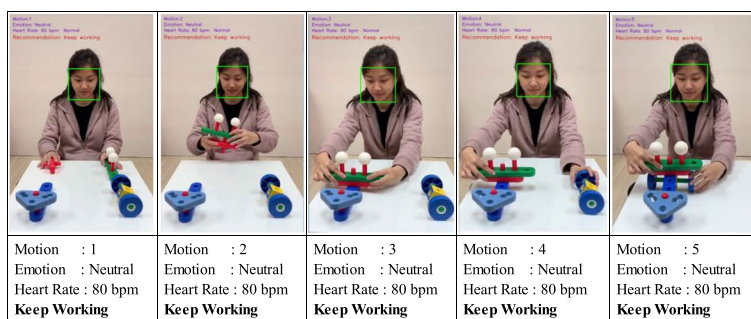
medical diagnostics, museum tutoring, and nursing. The proposed model shows a 7.8% of improvement compared to the existing deep learning algorithm.

Secondly, customer and employee emotions results have significant impacts on service encounters [28]. We have illustrated several examples in the service industry. A fast-food restaurant employee shows his enthusiasm for service, providing friendly and ethical assistance to customers and the expression of happiness is largely detected due to the expression of massive smile emotions. Since the service quality attributes, customer satisfaction, and trust affect customer loyalty to the fast-food restaurant, by using the proposed FER model to detect customer emotions in the service process, customer loyalty can be optimized. Further, the retailer service quality dimensions affect legitimacy, resulting in customer revisit intention. Throughout the proposed FER process, a video case has shown that consumers' emotions can be effectively captured and utilized to improve their on-site service experiences.

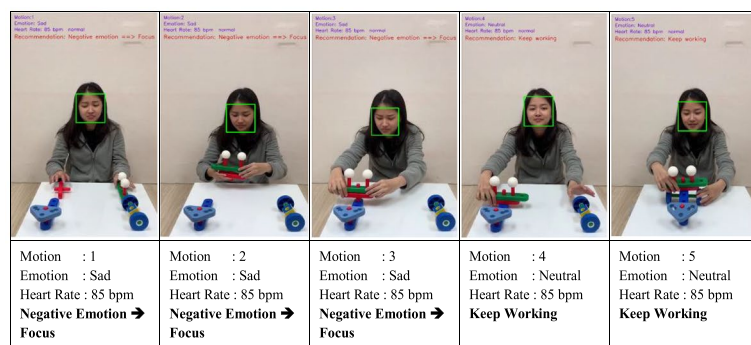
Thirdly, the FER model has been applied to the operation advice by monitoring the emotion when the player is assembling Lego, along with additional motion and heart rate monitoring functions. Accordingly, appropriate recommendations regarding focusing, keeping going, or stopping commands are provided. Such an application model also provides the potential to enable the safety function in the HMC domain. The model can monitor the emotions, motions, and vital signals of the human operator continuously.

The negative emotional response of customers can lead to service quality problems, and vice versa. To cope with negative emotions, a service remedy procedure can be implemented and triggered by integrating the proposed FER model to recognize the current emotional states of customers and advising the frontline employees to react. For instance, when the customer is experiencing negative emotions, our FER model suggests that the employee takes immediate action and provides appropriate services (including greetings, amusement and excitement services) to improve their emotional state. Moreover, advertisements for products/services often display menus with photographs of the product/service to evoke the kind of preferences of the customer to engender order placing [1, 48]. The proposed FER model will help observe constantly the customer's emotions and trigger a timely notification to the employee as necessary.

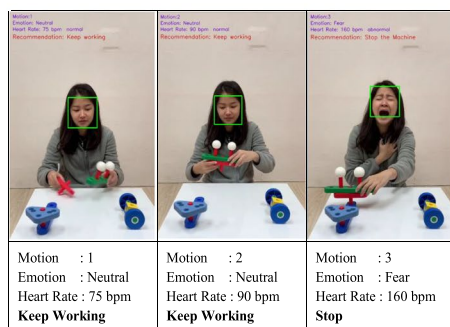
Alternatively, during a service process, unconscious but unpleasure emotions such as disgust, sadness, anger or even neutral of employees can initiate the emotional contagion



(a) Normal playing state.



(b) Abnormal emotion occurs.



(c) Abnormal heart rate/fear emotion occurs.

Fig. 11 Examples of Lego player emotional status

process, and customers are susceptible to catching employees' negative emotions through emotional contagion [49, 50]. The proposed FER model will remind the employee timely and get rid of further service failures.

In brief, the proposed FER model benefits from the two individual models. The MTCNN model is customized to properly align the face boundary. The other purpose of using MTCNN in the study is to remove blemishes on images. It also conducts the pre-processing of alignment and automatic labeling of images. Then, the YOLOv3 model for FER in service activities is developed to detect emotions in real time. The automatically marked emotional classification images of customers and servers are obtained.

6 Conclusions

In this study, we developed a two-stage model consisting of MTCNN and YOLOv3 algorithms for facial emotion recognition in the service industry. In the first stage, a customized MTCNN is developed to achieve face recognition and face alignment preprocessing. It solved the problem in which flawed images of a face cannot be detected in the training dataset and the problem of labeling images taking too much time. In the second phase, we implemented the YOLOv3 one-shot CNN architecture to increase the speed and achieve real-time FER. Results show that the proposed model performs well in detecting emotions of grayscale, flawed and low-pixel facial images.

The FER2013, RAF-DB, CK+, and mixed datasets are used as the model training set, and the performance of the proposed two-stage model is compared to YOLOv3. The precision of the proposed model is higher than that of the YOLOv3 model in training FER2013, CK+, and mixed datasets. By training our model using global test datasets (namely, RAF-DB, CK+, and FER2013), its accuracy reaches 86.1%, 95.6%, and 69%, respectively. The precision of the FER2013 dataset, CK+ dataset, and mixed dataset increased by 2%, 1%, and 2% respectively. Therefore, the two-stage model can increase the precision of the model compared with a single YOLOv3 model.

The proposed FER model is applied to the yawning behavior of car drivers and reaches an accuracy of 82%. In addition, using the short film in the AFEW data set as a model test, the precision of the proposed model is 5.9% higher than that of the YOLOv3 model, reaching an accuracy of 62.5%.

The contribution of this study is twofold. In terms of theoretical contribution, this study proposed a unified model that integrates two different CNNs. We developed a deep-learning based two-stage FER system that can effectively detect emotions to be consistent with Ekman's six basic emotions. Unorganized training images regarding facial expressions in the service industry can be filtered and automatically labeled through the first stage by using the proposed FER model. Manually labeled tasks and incorrectly marked positions due to subjective judgment are minimized. The second stage of the proposed FER model detects subtle facial objects precisely in real time, which facilitates FER in practice.

In terms of practical contribution, the proposed model can be deployed in real time to analyze human emotions in the service industry (such as smart retail and hypermarkets) to identify customer satisfaction with products and/or services, fatigue in working domains, and safety in jobs. Such facial recognition technology can motivate new digital business models and change customer-server dynamics. The proposed FER model will help improve customer relationships, provide pleasant transaction solutions, and even help to broaden product offerings and promotions. It can also be deployed in security, entertainment and hospitality, human resource management, education and other cases.

Future research can refine the optimization of YOLOv3 or use other algorithms for merging and comparison and to remove the influence of more external environmental factors. MTCNN and YOLOv3 are systems with deep learning architecture and require long computing time. Therefore, in the future, we can reduce the complexity of the FER system by applying the proposed FER model to actual production work and/or improving customer service. More thought and discussion on yawning and arousal and their relationship to the basic emotions, as well as how they enter decision-making, can be set as future research. A comparative experiment among those emerging YOLO versions and other transformer-based models, such as Vision Transformer (ViT) [52], also remains a future study.

Appendixes are available online

https://docs.google.com/document/d/1PAsd0mQ5BfEH2ZtK-wwZzOCF-zJVER0f/edit?usp=share_link&ouid=115521230954368008976&rtopof=true&sd=true

Data availability The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of interests/competing interests All the authors declare that they have no conflict of interest.

References

1. Chuah SHW, Yu J (2021) The future of service: the power of emotion in human-robot interaction. *J Retail Consum Serv* 61:102551
2. Shrivastava K, Kumar S, Jain DK (2019) An effective approach for emotion detection in multimedia text data using sequence based convolutional neural network. *Multimed Tools Appl* 78(20):29607–29639
3. Chang YC, Hsing YC (2021) Emotion-infused deep neural network for emotionally resonant conversation. *Appl Soft Comput* 113:107861
4. Bi Y, Xue B, Zhang M (2021) Multi-objective genetic programming for feature learning in face recognition. *Appl Soft Comput* 103:107152
5. Huang MH, Rust RT (2021) Engaged to a robot? The role of AI in service. *J Serv Res* 24(1):30–41
6. Poria S, Cambria E, Bajpai R, Hussain A (2017) A review of affective computing: from unimodal analysis to multimodal fusion. *Inf Fus* 37:98–125
7. Kwon S (2021) Att-net: enhanced emotion recognition system using lightweight self-attention module. *Appl Soft Comput* 102:107101
8. Ekman P, Friesen WV (1971) Constants across cultures in the face and emotion. *J Pers Soc Psychol* 17(2):124
9. Ghofrani A, Toroghi RM, Ghanbari S (2019) Realtime face-detection and emotion recognition using mtcnn and minishufflenet v2. In 2019 5th conference on knowledge based engineering and innovation (KBEI) (pp 817–821). IEEE
10. Su C, Wang G (2020) Design and application of learner emotion recognition for classroom. *J Phys Conf Ser* 1651(1):012158 IOP Publishing
11. Li X, Yang Z, Wu H (2020) Face detection based on receptive field enhanced multi-task cascaded convolutional neural networks. *IEEE Access* 8:174922–174930
12. Redmon J, Farhadi A (2018) YOLOv3: An Incremental Improvement. *arXiv pre-print server*. arxiv:1804.02767
13. Robinson S, Orsingher C, Alkire L, De Keyser A, Giebelhausen M, Papamichail KN, ... Temerak MS (2020) Frontline encounters of the AI kind: an evolved service encounter framework. *J Bus Res* 116:366–376
14. Prentice C, Nguyen M (2020) Engaging and retaining customers with AI and employee service. *J Retail Consum Serv* 56:102186
15. Hollebeek LD, Sprott DE, Brady MK (2021) Rise of the machines? Customer engagement in automated service interactions. *J Serv Res* 24(1):3–8
16. Henkel AP, Bromuri S, Iren D, Urovi V (2020) Half human, half machine—augmenting service employees with AI for interpersonal emotion regulation. *J Serv Manag* 31(2)
17. Ekman P, Friesen WV (2003) Unmasking the face: a guide to recognizing emotions from facial clues (Vol. 10). Ishk
18. Hamm J, Kohler CG, Gur RC, Verma R (2011) Automated facial action coding system for dynamic analysis of facial expressions in neuropsychiatric disorders. *J Neurosci Methods* 200(2):237–256
19. Bagozzi RP (2020a) Foundations of emotional research and its application. *Korean J Market* 35(4):1–51

20. Kotler P, Kartajaya H, Setiawan I (2021) Marketing 5.0: technology for humanity. John Wiley & Sons
21. Mehrabian A (2017) Nonverbal communication. Routledge
22. Ko BC (2018) A brief review of facial emotion recognition based on visual information. *Sensors* 18(2):401
23. Lei J, Han Q, Chen L, Lai Z, Zeng L, Liu X (2017) A novel side face contour extraction algorithm for driving fatigue statue recognition. *IEEE Access* 5:5723–5730
24. Yang B, Cao J, Ni R, Zhang Y (2017) Facial expression recognition using weighted mixture deep neural network based on double-channel facial images. *IEEE Access* 6:4630–4640
25. Kim BK, Roh J, Dong SY, Lee SY (2016) Hierarchical committee of deep convolutional neural networks for robust facial expression recognition. *J Multimod User Interfaces* 10(2):173–189
26. Huang R, Pedoeem J, Chen C (2018) YOLO-LITE: a real-time object detection algorithm optimized for non-GPU computers. In 2018 IEEE international conference on big data (big data) (pp 2503–2510). IEEE
27. Wang KJ, Asrini LJ, Sanjaya L, Nguyen PH (2022) Model for deep learning-based skill transfer in an assembly process. *Adv Eng Inform* 52:101643
28. Lee MC, Chiang SY, Yeh SC, Wen TF (2020) Study on emotion recognition and companion Chatbot using deep neural network. *Multimed Tools Appl* 79(27):19629–19657
29. Viola P, Jones M (2001) Rapid object detection using a boosted cascade of simple features. In proceedings of the 2001 IEEE computer society conference on computer vision and pattern recognition. CVPR 2001 (Vol. 1, pp I-I). IEEE
30. Zhang K, Zhang Z, Li Z, Qiao Y (2016) Joint face detection and alignment using multitask cascaded convolutional networks. *IEEE Signal Process Lett* 23(10):1499–1503
31. Casalboni A, Qin M, Sabo I, Hawkins A, Alapati S, Badola V, ..., Cecaro F (2017) Google Vision vs. Amazon Rekognition: A Vendor-Neutral Comparison. Cloud Academy
32. Evergreen (2021) Facial recognition services review <https://evergreen.team/articles/facial-recognition-services-comparison.html>. Accessed 10 May 2022
33. Goodfellow IJ, Erhan D, Carrier PL, Courville A, Mirza M, Hamner B, ... Bengio Y (2013) Challenges in representation learning: a report on three machine learning contests. In: International conference on neural information processing. Springer, Berlin, Heidelberg, pp 117–124
34. Li S, Deng W, Du J (2017) Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild. In proceedings of the IEEE conference on computer vision and pattern recognition (pp 2852–2861)
35. Lucey P, Cohn JF, Kanade T, Saragih J, Ambadar Z, Matthews I (2010) The extended cohn-kanade dataset (ck+): a complete dataset for action unit and emotion-specified expression. In 2010 IEEE computer society conference on computer vision and pattern recognition-workshops (pp 94–101). IEEE
36. Bochkovskiy A, Wang CY, Liao HYM (2020) Yolov4: optimal speed and accuracy of object detection. arXiv preprint arXiv:2004.10934
37. Giannopoulos P, Perikos I, Hatzilygeroudis I (2018) Deep learning approaches for facial emotion recognition: a case study on FER-2013. In: Advances in hybridization of intelligent methods. Springer, Cham, pp 1–16
38. Pang L, Liu H, Chen Y, Miao J (2020) Real-time concealed object detection from passive millimeter wave images based on the YOLOv3 algorithm. *Sensors* 20(6):1678
39. Dhall A, Goecke R, Lucey S, Gedeon T (2012) Collecting large, richly annotated facial-expression databases from movies. *IEEE MultiMed* 19(3):34–41
40. Kossaifi J, Tzimiropoulos G, Todorovic S, Pantic M (2017) AFEW-VA database for valence and arousal estimation in-the-wild, image and vision computing
41. Vielzeuf V, Pateux S, Jurie F (2017) Temporal multimodal fusion for video emotion classification in the wild. arXiv:1709.07200
42. Söderlund M, Rosengren S (2008) Revisiting the smiling service worker and customer satisfaction. *Int J Serv Ind Manag* 19(5):552–574
43. González-Rodríguez MR, Díaz-Fernández MC, Gómez CP (2020) Facial-expression recognition: an emergent approach to the measurement of tourist satisfaction through emotions. *Telematics Inform* 51:101404
44. Indira DNVSLS, Sumalatha L, Markapudi BR (2021) Multi facial expression recognition (MFER) for identifying customer satisfaction on products using deep CNN and Haar Cascade classifier. In IOP conference series: materials science and engineering (Vol. 1074, No. 1, p 012033). IOP Publishing
45. Singh G, Slack N, Sharma S, Mudaliar K, Narayan S, Kaur R, Sharma KU (2021) Antecedents involved in developing fast-food restaurant customer loyalty. *The TQM J*

46. Yang K, Kim J, Min J, Hernandez-Calderon A (2021) Effects of retailers' service quality and legitimacy on behavioral intention: the role of emotions during COVID-19. *Serv Ind J* 41(1–2):84–106
47. American Heart Association (2017) <https://www.heart.org/>. Accessed 10 May 2022
48. Small DA, Verrochi NM (2009) The face of need: facial emotion expression on charity advertisements. *J Mark Res* 46(6):777–787
49. Dallimore KS, Sparks BA, Butcher K (2007) The influence of angry customer outbursts on service providers' facial displays and affective states. *J Serv Res* 10(1):78–92
50. Perusquía-Hernández M (2021) Are people happy when they smile?: affective assessments based on automatic smile genuineness identification. *Emot Stud* 6(1):57–71
51. NVIDIA (2021) The NVIDIA transfer learning toolkit. https://docs.nvidia.com/tlt/tlt-user-guide/text/object_detection/yolo_v3.html. Accessed 10 May 2022
52. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Houlsby N (2020) An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*. Accessed 10 May 2022

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